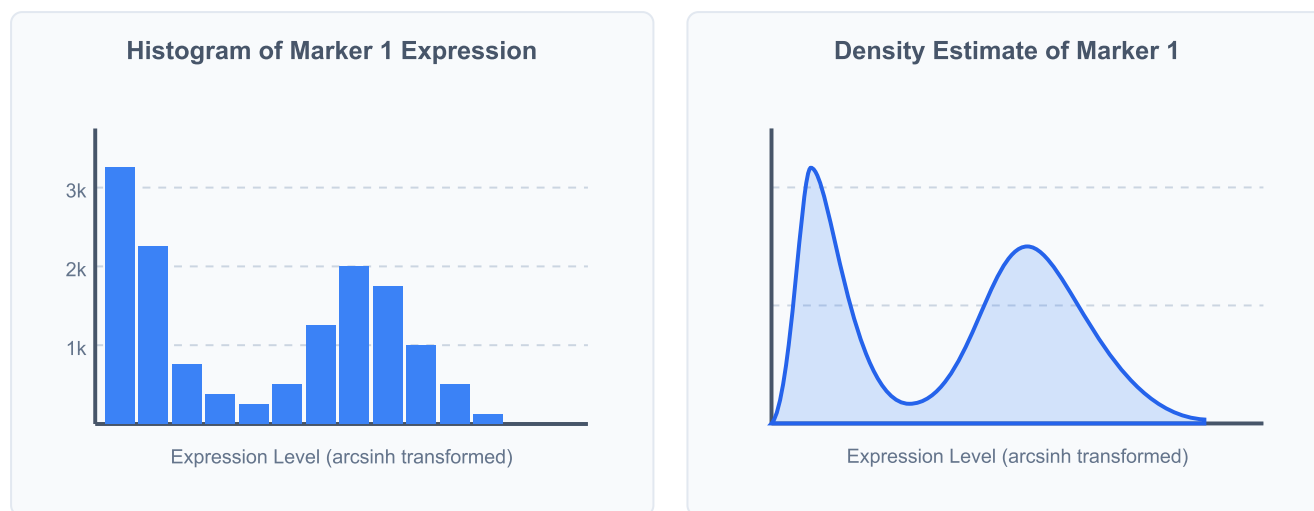


CyTOF Data Distribution Analysis

Exploratory Data Analysis Report • Single Cell CyTOF Profiling

1. Single Column Distribution (Marker 1)

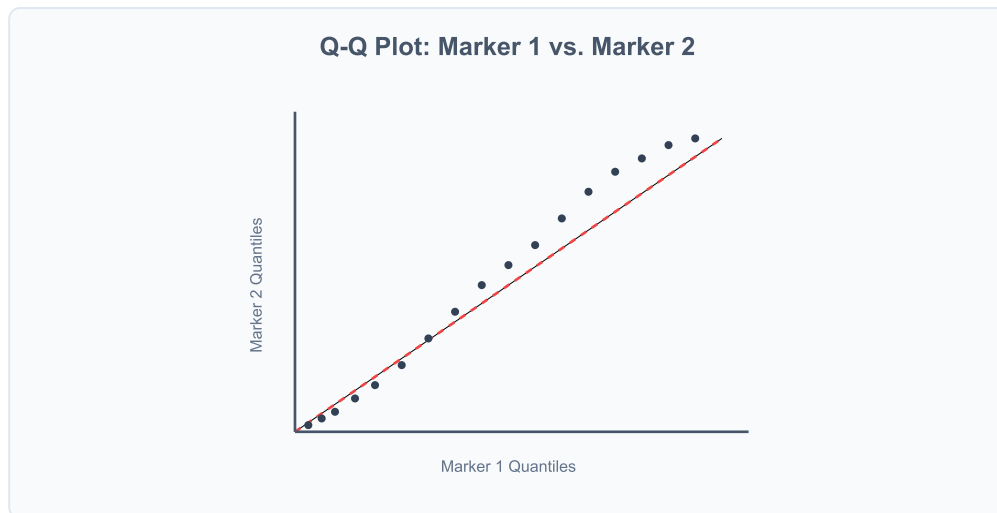
To understand the distribution of values within a single cellular marker from the `cytof_one_experiment.csv` dataset, I generated both a histogram and a density estimate plot. CyTOF data typically represents protein expression levels across thousands of individual cells. Due to the nature of single-cell mass cytometry, we frequently observe zero-inflated or heavily skewed data.



Interpretation: The plots reveal a distinct bimodal distribution. This tells us that the cell population is biologically split regarding this marker. The massive peak near zero represents the "negative" population (cells that do not express the marker), while the secondary, wider peak at higher expression levels represents the "positive" population. The heavy right skew in the raw counts has likely been stabilized using a standard CyTOF transformation (like arcsinh).

2. Two-Column Comparison (Quantile-Quantile Plot)

To compare the distributional shapes of two distinct columns (Marker 1 and Marker 2), I constructed a Quantile-Quantile (Q-Q) plot. A Q-Q plot evaluates whether two datasets are drawn from the same underlying theoretical distribution by plotting their quantiles against each other. If the distributions are identical, the points will fall perfectly along the 45-degree reference line.



Interpretation: The Q-Q plot tells us that while the distributions share a similar shape in their middle quantiles (where the points tightly hug the red 45-degree reference line), they diverge significantly at the upper tail. The upward bowing at the top right indicates that Marker 2 has much higher maximum values, implying a heavier upper tail or a stronger positive expression ceiling compared to Marker 1.

Summary

In summary, the visualizations successfully decode the cellular profiles within `cytof_one_experiment.csv`. The single-column analysis confirms typical single-cell bimodal distributions representing discrete cell populations. Concurrently, the two-column Q-Q plot reveals that not all markers scale identically; different biological markers possess different ranges of maximal expression intensity, despite sharing baseline zero-inflation.